Random Processes

Chapter 3: Practice problems (Lectures 8–9)

University of Michigan

Scope. Exit probabilities and expected exit times, the matrix form $(I-Q)^{-1}$, gambler's ruin, and countable state spaces (positive, null recurrence and transience). Notation follows the notes: $V_A = \inf\{n \geq 0 : X_n \in A\}$ counts time 0, $T_y = \inf\{n \geq 1 : X_n = y\}$ does not. A $\star$ on a problem or a part means it is harder than the Fall 2026 practice midterm; unmarked problems are at about exam level. Solutions start on a new page after the problems.

Problems

Problem 1 (Random walk on a graph).
A walker moves on the graph with vertices $\{1, ..., 5\}$ and edges

$$\{1,2\}, \{1,3\}, \{2,3\}, \{2,4\}, \{3,4\}, \{3,5\}, \{4,5\},$$

choosing a neighbour uniformly at random at each step. It starts at 1.

(a) Find the probability of reaching 5 before 4.

(b) Find the expected number of steps until it first reaches $\{4, 5\}$.

(c) Find the expected number of steps to return to 1. Use a method different from (a)–(b).

(d) A step from i to j costs $|i-j|$. Write (do not solve) a system of equations for the expected total cost until the first arrival at 5, including the final step.

Problem 2 (Gambler's ruin numbers).

(a) $N = 5$, $p = 0.4$, initial capital $x = 2$. Find the probability of reaching 5 before 0 and the expected duration.

(b) At roulette ($p = \frac{18}{38}$) you start with \$10 and stop at \$20 or \$0, betting \$1 each time. What is the probability of reaching \$20? Compare with a fair game.

(c) Same game, but you are allowed to keep playing forever if you never hit \$0 (no upper target). What is the probability you are never ruined?

Problem 3 (Gambler's ruin with ties).
On $\{0, ..., N\}$ with 0, N absorbing, let $p(x, x+1) = p$, $p(x, x-1) = q$, $p(x, x) = s$ for $0 < x < N$, where $p, q > 0$ and $p + q + s = 1$.

(a) Show that $h(x) = \mathbb{P}_x(V_N < V_0)$ is the same as for the walk without ties with up-probability $p' = \frac{p}{p+q}$.

(b) Show that the expected duration is $g(x) = g'(x)/(1-s)$, where g' is the duration for the walk without ties with up-probability p'.

(c) Evaluate both for $N = 4$, $x = 2$, $p = q = \frac{1}{4}$, $s = \frac{1}{2}$.

Problem 4 (Fundamental matrix: a two-year program).
A student in year 1 repeats the year with probability 0.1, advances to year 2 with probability 0.8 and drops out with probability 0.1. In year 2 the student repeats with probability 0.1, graduates with probability 0.85 and drops out with probability 0.05. Graduate and Dropout are absorbing.

(a) Write Q (transient part) and compute $F = (I - Q)^{-1}$. Interpret each entry.

(b) Starting in year 1, find the expected number of years in the program.

(c) Starting in year 1, find the probability of graduating.

Problem 5 (Waiting for patterns).
A coin shows heads with probability p; flips are independent.

(a) For $p = \frac{1}{2}$, find the expected number of flips until HH first appears, and until HT first appears.

(b) For general p, find the probability that HH appears before TT. Evaluate at $p = \frac{2}{3}$.

Problem 6 (Reflection at zero).
Simple symmetric walk on $\{0, 1, ..., N\}$ with $p(0, 1) = 1$ (reflection at 0) and N absorbing.

(a) Write the equations for $g(x) = \mathbb{E}_x V_N$.

(b) Verify that $g(x) = N^2 - x^2$ solves them, and explain why this is the answer.

(c) Compare $g(0)$ with the expected duration $x(N - x)$ of the fair gambler's ruin. Which is larger at $x = N/2$, and why?

Problem 7 (A discrete queue).
On $S = \mathbb{Z}_{\geq 0}$ let $p(i, i + 1) = p$ for $i \geq 0$, $p(i, i - 1) = q$ for $i \geq 1$, $p(i, i) = 1 - p - q$ for $i \geq 1$, and $p(0, 0) = 1 - p$, where $p, q > 0$ and $p + q \leq 1$.

(a) Show the chain is irreducible and aperiodic.

(b) Find all p, q for which a stationary distribution exists, and find it.

(c) In that case, find $\mathbb{E}_0 T_0$ and $\lim_n P^n(5, 0)$.

(d) For $p = q$, is the chain transient, null recurrent or positive recurrent?

Problem 8 (Renewal chains: three behaviours).
On $S = \mathbb{Z}_{\geq 0}$ let $p(i, i + 1) = a_i$ and $p(i, 0) = 1 - a_i$, with $0 < a_i < 1$.

(a) Show that $\mathbb{P}_0(T_0 > k) = a_0 a_1 \cdots a_{k-1}$ for $k \geq 1$.

(b) Classify the chain (transient / null recurrent / positive recurrent) for

$$\text{(i) } a_i = \frac{1}{i+2}, \quad \text{(ii) } a_i = \frac{i+1}{i+2}, \quad \text{(iii) } a_i = 1 - \frac{1}{(i+2)^2}.$$

(c) In case (i), find $\mathbb{E}_0 T_0$ and the stationary distribution.

(d) $\star$ In case (iii), find $\mathbb{P}_0(T_0 = \infty)$ exactly.

Problem 9 ⋆ (Mean hitting time on a half-line).
Let $p < q$, $p + q = 1$, and consider the walk on $\mathbb{Z}_{\geq 0}$ with $p(x, x+1) = p$, $p(x, x-1) = q$ for $x \geq 1$, and 0 absorbing.

(a) Using the gambler's ruin duration $g_N(x) = \mathbb{E}_x(V_0 \wedge V_N)$ and monotone convergence, show $\mathbb{E}_x V_0 = \frac{x}{q-p}$.

(b) What goes wrong when $p = q$?

Problem 10 (True or false; justify briefly).

(a) In an irreducible chain, if one state is null recurrent then all states are.

(b) An irreducible recurrent chain on a countably infinite space always has a stationary distribution.

(c) If $\mathbb{E}_x T_x < \infty$ for one state x of an irreducible chain, a stationary distribution exists.

(d) On a finite S, $h(x) = \mathbb{P}_x(V_A < V_B)$ is always the unique solution of the boundary-value equations.

(e) For simple symmetric random walk on $\mathbb{Z}$, $\mathbb{E}_1 V_0 < \infty$.

(f) A transient state y has $\mathbb{E}_y T_y = \infty$.

(g) On a finite S, if every state of $D = S \setminus A$ can reach A, then $I - Q$ is invertible, where $Q = P|_{D \times D}$.

Solutions

Solution 1.
Degrees: $d(1)=2$, $d(2)=3$, $d(3)=4$, $d(4)=3$, $d(5)=2$; $2|E|=14$.

(a) $h(x)=\mathbb{P}_x(V_5<V_4)$: $h(5)=1$, $h(4)=0$ and

$$h(1)=\frac{h(2)+h(3)}{2},\quad h(2)=\frac{h(1)+h(3)+0}{3},\quad h(3)=\frac{h(1)+h(2)+0+1}{4}.$$

The solution is $h(1)=\frac{4}{13}$, $h(2)=\frac{3}{13}$, $h(3)=\frac{5}{13}$. So the answer is $\frac{4}{13}$.

(b) $g(x)=\mathbb{E}_x V_{\{4,5\}}$: $g(4)=g(5)=0$ and

$$g(1)=1+\frac{g(2)+g(3)}{2},\quad g(2)=1+\frac{g(1)+g(3)}{3},\quad g(3)=1+\frac{g(1)+g(2)}{4},$$

giving $g(1)=\frac{53}{13}$, $g(2)=\frac{43}{13}$, $g(3)=\frac{37}{13}$. So the answer is $\frac{53}{13}\approx 4.08$.

(c) The graph is connected, so the walk is irreducible with $\pi(u)=d(u)/(2|E|)$. By Theorem 6.7(3), $\mathbb{E}_1 T_1=1/\pi(1)=\frac{14}{2}=7$.

(d) Let $c(x)$ be the expected cost until V_5. Then $c(5)=0$, and for $x\neq 5$, conditioning on the first step,

$$c(x)=\sum_{y\sim x}\frac{1}{d(x)}(|x-y|+c(y)).$$

Explicitly:

$$\begin{aligned}
c(1)&=\frac{1}{2}(1+c(2))+\frac{1}{2}(2+c(3)),\\
c(2)&=\frac{1}{3}(1+c(1))+\frac{1}{3}(1+c(3))+\frac{1}{3}(2+c(4)),\\
c(3)&=\frac{1}{4}(2+c(1))+\frac{1}{4}(1+c(2))+\frac{1}{4}(1+c(4))+\frac{1}{4}(2+0),\\
c(4)&=\frac{1}{3}(2+c(2))+\frac{1}{3}(1+c(3))+\frac{1}{3}(1+0).
\end{aligned}$$

Solution 2.
(a) $r=\frac{q}{p}=1.5$. $h(2)=\frac{1-r^2}{1-r^5}=\frac{40}{211}\approx 0.190$ and $g(2)=\frac{2-5h(2)}{q-p}=\frac{1110}{211}\approx 5.26$ bets.

(b) $r=\frac{20}{18}=\frac{10}{9}$: $h(10)=\frac{1-r^{10}}{1-r^{20}}\approx 0.259$. In a fair game it is $\frac{10}{20}=0.5$: a small house edge halves your chance.

(c) With no upper target, the probability of never being ruined is $\lim_{N\to\infty} h_N(10)$. Since $p<q$, this limit is 0 (Proposition 8.17): ruin is certain.

Solution 3.
(a) For $0<x<N$, $h(x)=ph(x+1)+qh(x-1)+sh(x)$, i.e. $(1-s)h(x)=ph(x+1)+qh(x-1)$. Divide by $1-s=p+q$:

$$h(x)=p'h(x+1)+q'h(x-1),\quad p'=\frac{p}{p+q},\ q'=\frac{q}{p+q}.$$

These are the equations without ties, with the same boundary values; by uniqueness h is the no-ties answer, with ratio $\frac{q'}{p'} = \frac{q}{p}$.

(b) $(1-s)g(x) = 1 + pg(x+1) + qg(x-1)$. Multiply $g'(x) = 1 + p'g'(x+1) + q'g'(x-1)$ by $\frac{1}{1-s}$ to see that $\frac{g'}{1-s}$ solves the same equations with zero boundary values; by uniqueness $g = \frac{g'}{1-s}$. (Each "real" move takes on average $\frac{1}{1-s}$ steps.)

(c) $p' = q' = \frac{1}{2}$, so $h(2) = \frac{2}{4} = \frac{1}{2}$ and $g(2) = 2(4-2)/\left(\frac{1}{2}\right) = 8$.

Solution 4.

(a) With transient states (year 1, year 2),

$$Q = \begin{pmatrix} 0.1 & 0.8 \\ 0 & 0.1 \end{pmatrix}, \quad I - Q = \begin{pmatrix} 0.9 & -0.8 \\ 0 & 0.9 \end{pmatrix}, \quad F = (I-Q)^{-1} = \begin{pmatrix} \frac{10}{9} & \frac{80}{81} \\ 0 & \frac{10}{9} \end{pmatrix}.$$

$F(x, y)$ is the expected number of years spent in year y starting from year x (counting the starting year). For example $F(1, 2) = \frac{80}{81}$: a year-1 student spends on average $\frac{80}{81}$ years in year 2.

(b) $g = F\mathbf{1}$: $g(1) = \frac{10}{9} + \frac{80}{81} = \frac{170}{81} \approx 2.10$ years.

(c) Let $b = (0, 0.85)^\top$ be the one-step graduation probabilities. Then $h = Fb$, so $h(1) = \frac{80}{81} \cdot 0.85 = \frac{68}{81} \approx 0.840$. (From year 2: $h(2) = \frac{10}{9} \cdot 0.85 = \frac{17}{18}$.)

Solution 5.

(a) Track the relevant suffix. For HH with states $\emptyset$ (start or last flip T) and H: $a = 1 + \frac{a}{2} + \frac{b}{2}$ and $b = 1 + \frac{a}{2}$, where a, b are the expected remaining flips. Hence $a = 6$. For HT: $a = 1 + \frac{a}{2} + \frac{b}{2}$ and $b = 1 + \frac{b}{2}$ (from H, a T finishes and an H keeps us at H), so $b = 2$ and $a = 4$. The answers are 6 and 4: HT comes sooner because a failed attempt at HT (another H) does not lose progress.

(b) States: last flip H or T. Let $u = \mathbb{P}(HH \text{ first} \mid \text{last} = H)$ and $v = \mathbb{P}(HH \text{ first} \mid \text{last} = T)$. Then

$$u = p + qv, \quad v = pu \quad \Longrightarrow \quad u = \frac{p}{1-pq}, \quad v = \frac{p^2}{1-pq}.$$

From the start, $pu + qv = \frac{p^2(1+q)}{1-pq}$. At $p = \frac{2}{3}$: $\frac{\frac{4}{9} \cdot \frac{4}{3}}{\frac{7}{9}} = \frac{16}{21}$.

Solution 6.

(a) $g(N) = 0$, $g(0) = 1 + g(1)$, and $g(x) = 1 + \frac{g(x+1)+g(x-1)}{2}$ for $0 < x < N$.

(b) For $0 < x < N$: $1 + \frac{(N^2-(x+1)^2)+(N^2-(x-1)^2)}{2} = 1 + N^2 - x^2 - 1 = N^2 - x^2$. At 0: $1 + N^2 - 1 = N^2$. At N: 0. Every state reaches N, so the solution is unique (Theorem 8.11) and $g(x) = N^2 - x^2$.

(c) At $x = N/2$: $N^2 - \frac{N^2}{4} = 3N^2/4$ versus $N^2/4$. Reflection removes the "escape" at 0, so the walk must eventually go all the way to N: three times as long.

Solution 7.

(a) Neighbours communicate in both directions, so the chain is irreducible; $p(0, 0) = 1 - p > 0$ gives aperiodicity.

(b) Detailed balance $\pi(i)p = \pi(i+1)q$ gives $\pi(i) = \pi(0)\left(\frac{p}{q}\right)^i$. This is normalisable iff $p < q$, and then $\pi(i) = \left(1 - \frac{p}{q}\right)\left(\frac{p}{q}\right)^i$. Since the chain is irreducible, a stationary distribution exists iff it is positive recurrent; for $p \geq q$, the same truncation argument as for the reflecting walk (Theorem 9.5) shows it is null recurrent ($p = q$) or transient ($p > q$), so none exists.

(c) $\mathbb{E}_0 T_0 = 1/\pi(0) = \frac{q}{q-p}$, and by Theorem 6.7(1), $\lim_n P^n(5,0) = \pi(0) = 1 - \frac{p}{q}$.

(d) Null recurrent: the holding probabilities only slow the walk down (each real move takes a geometric number of steps), so the embedded walk is the symmetric reflecting walk of Theorem 9.5(b).

Solution 8.

(a) Starting at 0, $T_0 > k$ iff the first k steps are all "up", i.e. $0 \to 1 \to \cdots \to k$, which has probability $a_0 a_1 \cdots a_{k-1}$.

(b) Recall $\mathbb{P}_0(T_0 = \infty) = \lim_k \mathbb{P}_0(T_0 > k)$ and $\mathbb{E}_0 T_0 = \sum_{k \geq 0} \mathbb{P}_0(T_0 > k)$.

- (i) $\mathbb{P}_0(T_0 > k) = \frac{1}{(k+1)!} \to 0$ and $\sum_k \frac{1}{(k+1)!} = e - 1 < \infty$: positive recurrent.
- (ii) $\mathbb{P}_0(T_0 > k) = \frac{1}{2} \cdot \frac{2}{3} \cdots \frac{k}{k+1} = \frac{1}{k+1} \to 0$, but $\sum_k \frac{1}{k+1} = \infty$: null recurrent.
- (iii) $\mathbb{P}_0(T_0 > k) = \prod_{j=2}^{k+1}\left(1 - \frac{1}{j^2}\right) \to \frac{1}{2} > 0$ (see (d)): transient.

The chain is irreducible, so the type of 0 is the type of every state.

(c) $\mathbb{E}_0 T_0 = e - 1$. Using the cycle measure, $\pi(i) \propto \mathbb{P}_0(T_0 > i) = \frac{1}{(i+1)!}$ (the only possible visit to i in an excursion is at time i), so

$$\pi(i) = \frac{1}{(e-1)(i+1)!} \quad (i \geq 0).$$

Check: $\pi(0) = \frac{1}{e-1} = 1/\mathbb{E}_0 T_0$.

(d) Telescoping, $\prod_{j=2}^{n}\left(1 - \frac{1}{j^2}\right) = \prod_{j=2}^{n} \frac{(j-1)(j+1)}{j^2} = \frac{n+1}{2n} \to \frac{1}{2}$. So $\mathbb{P}_0(T_0 = \infty) = \frac{1}{2}$.

Solution 9.

(a) For $N > x$, the walk agrees with gambler's ruin on $\{0, ..., N\}$ until $V_0 \wedge V_N$, so $\mathbb{E}_x(V_0 \wedge V_N) = g_N(x) = \frac{x - N h_N(x)}{q-p}$, where $h_N(x) = \frac{r^x - 1}{r^N - 1}$ and $r = \frac{q}{p} > 1$. Since $N h_N(x) \to 0$, $g_N(x) \to \frac{x}{q-p}$. Upward steps have size one, so $V_N \geq N - x \to \infty$, hence $V_0 \wedge V_N \uparrow V_0$, and monotone convergence gives $\mathbb{E}_x V_0 = \lim_N g_N(x) = \frac{x}{q-p}$.

(b) For $p = q$, $g_N(x) = x(N - x) \to \infty$, so $\mathbb{E}_x V_0 = \infty$ even though $V_0 < \infty$ almost surely: this is null recurrence.

Solution 10.

(a) True: Corollary 9.7 (trichotomy).

(b) False: simple symmetric random walk on $\mathbb{Z}$ is irreducible and recurrent but null recurrent, so it has no stationary distribution.

(c) True: Theorem 9.6, (1) $\Rightarrow$ (3), via the cycle measure.

(d) False: with an inaccessible closed class (for example, $P = I$ on three states) the equations leave h undetermined there. Uniqueness needs every state of D to reach $A \cup B$.

(e) False: $\mathbb{E}_1 V_0 \geq \mathbb{E}_1(V_0 \wedge V_N) = N - 1$ for every N.

(f) True: $\mathbb{P}_y\big(T_y = \infty\big) > 0$, so the expectation is infinite.

(g) True: Lemma 8.2 gives $\mathbb{P}_x(V_A > n) \to 0$ geometrically, so $Q^n \to 0$ and $\sum_n Q^n$ converges to $(I - Q)^{-1}$.